



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Product name: buprofezin 25% WP

Product use: Insecticide

Relevant identified uses of the substance/mixture and uses advised against:

Used as an insect growth regulator.

2. HAZARD IDENTIFICATION

Emergency Overview – CAUTION

Harmful if swallowed, inhaled, or absorbed through the skin.

Cause mild eye irritation.

Product is a white powder.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name (IUPAC): 2-*tert*-butylimino-3-isopropyl-5-phenylperhydro-1,3,5-thiadiazin-4-one

Common name: buprofezin

CAS Number: 69327-76-0

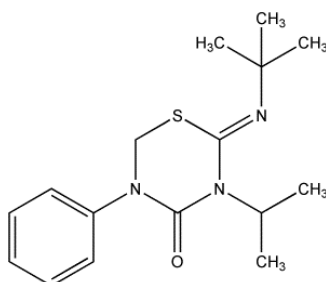
Ingredients and composition: Active ingredient – buprofezin 25% WP

Chemical family: Thiadiazine derivative

Formula: $C_{16}H_{23}N_3OS$

Molecular weight: 305.44 g/mol

Structure formula:





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4. FIRST AID MEASURES

Description of first aid measures

In case of swallowed:

Call a physician or the Poison Control Center. Drink one or two glasses of water. Induce vomiting by touching the back of the throat with the finger, or if available, by administering syrup of ipecac. If the person is unconscious, do not give anything by mouth and do not induce vomiting.

In case of eyes contact:

Flush eyes with plenty of water. Call a physician if irritation persists.

In case of skin contact:

Wash with plenty of soap and water. Get medical attention if irritation persists.

In case of Inhaled contact:

Remove the victim to fresh air. If not breathing, give the victim artificial respiration, preferably mouth-mouth. Get medical attention.

Hint for the Physician

Provide supportive care and symptomatic treatment (eye and respiratory irritation).

5. FIRE FIGHTING MEASURES

Flammable properties:

Does not flash (Flash point)

Auto ignition temperature:

Not combustible

Extinguishing Media:

Alcohol-resistant foam, carbon dioxide, dry chemicals, and water spray.

Hazardous Combustion Products:

Carbon dioxide, carbon monoxide, nitrogen oxides, and sulfur dioxide.

Special Fire Fighting Procedure:

Fireman should wear positive-pressure, self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Procedure for spill:

Do not breathe the dust.

Use a suitable dust respirator.

Avoid contact with skin and eyes.

Carefully sweep up the spilled product, avoiding formation of a dust cloud, and place it in a suitable container.



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Seal the container.

The area can be washed with water to remove the last traces of the product, but keep out of watercourses or sewers.

Inform authorities immediately if contamination occurs.

7. HANDLING AND STORAGE

Safe Handling:

Follow the manufacturer's instructions for cleaning and maintaining personal protective equipment (PPE). If no such instruction is provided for washables, use detergent and hot water.

Keep PPE separate from other laundry; wash it separately.

Storage Conditions:

Store in a cool and dry place.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering controls

Personal Protective Equipment:

Applicators and Handlers of Agricultural Products

All users of agricultural products that are within the scope of the EPA Worker Protection Standards (WPS) (40 CFR Part 170) must refer to the statements below or the product label for WPS-specified.

PPE, restricted entry interval (REI), and other precautionary statements.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Crystalline solid
Color:	White
Odor:	None
Melting/ Freezing Point:	104.2 – 105.5 °C
Solubility in water:	0.382 mg/L at 25°C
Solubility in organic solvents:	520 g/L in chloroform
	320 g/L in toluene
	240 g/L in acetone
	220 g/L in ethyl acetate
	20 g/L in methanol
	20 g/L in n-hexane



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Density: 1.183 – 1.187 g/ml at 25 °C
Vapor pressure: 0.0168 Pa (52 °C), 0.163 Pa (72 °C), 2.01 Pa (101 °C), 8.79 Pa (122 °C), 54.8 Pa (151 °C)
Octanol/water partition coefficient: log POW = 4.31 at °C

10. STABILITY AND REACTIVITY

This product is stable. The active ingredient is stable under dark and cool storage conditions; it is also stable in acid and alkali.

11. TOXICOLOGICAL INFORMATION

Information on toxicological effects

Acute toxicity LOW TOXICITY

LD50 (oral): 2,198 mg/kg (rat)
LD50 (dermal): > 5,000 mg/kg (rat)

12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, buprofezin:

Fish – Acute toxicity:

LC50 (96 h): > 260 µg/L (static conditions)
> 330 µg/L (flow-through conditions)

No chronic 24 aquatic toxicology studies were presented. The data package presented for the assessment of aquatic toxicity was limited.

Birds – Acute toxicity:

Bobwhite quail,

LD50: > 2,000 mg/kg
LC50 (5 days): > 5,243 mg/kg (dietary)

Mallard ducks,

LD50: > 2,000 mg/kg
LC50 (5 days): > 5,243 mg/kg (dietary)

Honey Bee – Acute toxicity:

LD50: < 100 µg/bee



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Earthworm – Acute toxicity:

Silkworms,

LC50: 1,000 – 2,000 ppm

Earthworms,

LC50 (14 days): > 1,000 mg/kg

LC50: 1,000 mg/kg (mortality)

13. DISPOSAL CONSIDERATIONS

Waste resulting from the use of this product may be disposed of on-site at an approved waste disposal facility.

Waste disposal information:

Do not reuse empty containers unless they are specifically designed to be re-filled. Empty container retains product residue. Dispose of empty containers in accordance with local regulations. Consult provincial environment ministry for advice on waste disposal. Industrial/commercial waste may be handled at licensed facilities only. Waste shipments must be securely packaged and properly labelled. Only licensed carriers may be used, and proper documents must accompany the shipment.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN No: 2588

Dangerous Goods Class: 6.1

Packaging Group: III

Marine pollutant: Yes

Other: Pesticide based on organophosphate, solid, toxic.

15. REGULATORY INFORMATION

Risk Phrases:

R36: Irritating to eyes.

R52: Harmful to aquatic organisms.

R55: Toxic to fauna.

Safety Phrases:



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S2: Keep out of reach of children

S13: Keep away from food, drink and animal feeding stuff

S20/21: When using do not eat, drink or smoke.

S24/25: Avoid contact with skin and eyes.

S36/37: Wear suitable protective clothing and gloves.

S51: Use only in well-ventilated areas.

S60: This material and its container must be disposed of as hazardous waste.

16. OTHER INFORMATION

The above information is believed to be correct based on our present knowledge but does not purport to be complete. For research use only by trained personnel. The burden of safe use of this material rests entirely with the user.